A twist-free converse theorem at level 36

27 October 2026

Abstract

We prove the holomorphic Conrey–Farmer converse theorem at level 36, for every positive even weight and either functional-equation sign. Only the local Euler conditions at $2, 3, 5$ are needed. For the positive Fricke eigenvalue, the conditions at $2, 3$ alone suffice. The proof uses the Fourier support forced by the ramified Euler factors to construct a three-dimensional module for the full modular group. A single $m = 5$ relation eliminates its possible noncongruence part. No twisted functional equations or additional coefficient estimates are assumed.

1 Statement and conventions

Conrey–Farmer [1] establish their holomorphic converse theorem for $5 \le N \le 12$, $14 \le N \le 17$, and $N = 23$. Conrey–Farmer–Odgers–Snaith [2] supply $N = 13$, and Bedert–Cooper–Oliver–Zhang [3] supply $N = 18, 20, 24$. The theorem below adds $N = 36$ to this list.

Theorem 1. *Let $k \ge 2$ be even. Let $a_1 = 1$, and suppose*

$$F(s) = \sum_{n \ge 1} \frac{a_n}{n^s}$$

converges in a right half-plane. Assume

1. *$\Lambda(s) = (6/(2\pi))^s \Gamma(s) F(s)$ extends to an entire function, bounded in every closed vertical strip, and*
$$\Lambda(s) = \omega \Lambda(k - s), \qquad \omega \in \{1, -1\};$$
2. *$a_n = 0$ whenever $(n, 6) > 1$;*
3. *for all $n \ge 1$,*
$$a_{5n} = a_5 a_n - 5^{k-1} a_{n/5},$$
where $a_{n/5} = 0$ if $5 \nmid n$.

Then

$$f(z) = \sum_{n \ge 1} a_n e^{2\pi i n z} \in S_k(\Gamma_0(36)).$$

If $(-1)^{k/2} \omega = 1$, hypothesis (3) is unnecessary.

The local hypotheses in the theorem follow from the partial Euler product

$$F(s) = \frac{1}{1 - a_5 5^{-s} + 5^{k-1-2s}} \sum_{(n,30)=1} \frac{a_n}{n^s}.$$

In particular, they follow from the full Conrey–Farmer Euler product at level 36: its factors at 2 and 3 are 1. No condition is imposed here at any other prime.

For $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2^+(\mathbb{R})$, use the normalized weight-k action

$$(v|_k M)(z) = (\det M)^{k/2}(cz+d)^{-k} v\left(\frac{az+b}{cz+d}\right).$$

We suppress the subscript k. Throughout k is positive and even, so nonzero scalar matrices act trivially and $-I$ acts trivially. Set

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \beta(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad H_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}.$$

Convergence in a right half-plane implies $a_n = O(n^C)$ for some $C \geq 0$: choose a real point of convergence, use that its summands are bounded, and increase the exponent if necessary. Thus the Fourier series defines a holomorphic function on $\mathfrak{H}$. The usual Mellin-inversion argument, as in [1, Section 1], gives

$$f|H_{36} = \varepsilon f, \qquad \varepsilon = (-1)^{k/2}\omega \in \{1, -1\}. \tag{1}$$

We distinguish the analytic root number ω from the Fricke eigenvalue ε to avoid a weight-dependent sign ambiguity.

2 Two elementary lemmas

Lemma 2 (Positive-weight elliptic vanishing)**.** *Let v be holomorphic on $\mathfrak{H}$, and let $k > 0$ be even. If $v|E = v$ for an infinite-order elliptic element $E \in \mathrm{SL}_2(\mathbb{R})$, then $v = 0$.*

Proof. Let $a \in \mathfrak{H}$ be the fixed point. Use the disk coordinate $w = (z-a)/(z-\bar{a})$ and write $m = k/2$. The holomorphic differential $v(z)(dz)^m$ has the form $V(w)(dw)^m$, with V holomorphic on the disk. The action of E is $w \mapsto \zeta w$, where $|\zeta| = 1$ and ζ has infinite order. Invariance says $V(w) = \zeta^m V(\zeta w)$. If $V(w) = \sum_{j \geq 0} c_j w^j$, then $(1 - \zeta^{j+m})c_j = 0$. Since $j + m > 0$, every coefficient vanishes. □

Lemma 3 (A congruence-kernel criterion)**.** *Let V be a complex right $\mathrm{PSL}_2(\mathbb{Z})$-module such that T^3 acts as $-\mathrm{Id}_V$. Then $\Gamma(6)$ acts trivially on V.*

Proof. Let χ_2 be the sign of the permutation induced by reduction modulo 2 on $\mathbb{P}^1(\mathbb{F}_2)$. Thus $\chi_2(S) = \chi_2(T) = -1$, and χ_2 is trivial on $\Gamma(2)$. Twist the action by χ_2. In the twisted action, the operators corresponding to S, T satisfy

$$S^2 = 1, \qquad (ST)^3 = 1, \qquad T^3 = 1.$$

The quotient of $\mathrm{PSL}_2(\mathbb{Z})$ obtained by imposing $T^3 = 1$ is the triangle group $(2, 3, 3)$, of order 12, and reduction modulo 3 identifies it with $\mathrm{PSL}_2(\mathbb{F}_3) \simeq A_4$. For completeness, this is the rotational symmetry group of a tetrahedron: its stabilizers at an edge midpoint, a vertex, and a face center have orders $2, 3, 3$. Equivalently, its standard triangle presentation gives the above relations and order 12. Reduction modulo 3 is surjective onto a group of that same order, so its kernel is exactly the normal closure of T^3 in the projective modular group. Consequently the twisted action is trivial on $\Gamma(3)$. On $\Gamma(6) \subseteq \Gamma(2) \cap \Gamma(3)$ the twist itself is trivial, proving the assertion. □

3 The three-dimensional module

Rescale the function by setting

$$h(z) = f(z/6).$$

Equation (1) gives $h|S = \varepsilon h$. Since the only possible Fourier indices of f are congruent to 1 or 5 modulo 6, every frequency of h has translation eigenvalue a primitive sixth root. Therefore

$$h|(T^2 - T + 1) = 0, \qquad h|T^{-1} = h - h|T, \qquad h|T^3 = -h. \tag{2}$$

Define

$$g = h|T, \qquad v = g|S.$$

Using $STS = T^{-1}ST^{-1}$, we have

$$\varepsilon v = h|STS = h|T^{-1}ST^{-1} = (\varepsilon h - v)|T^{-1}.$$

Right-multiplication by T yields $v|T = h - \varepsilon v$. Hence

	$\|S$	$\|T$
h	εh	g
g	v	$g - h$
v	g	$h - \varepsilon v$

(3)

In particular, $V = \operatorname{span}_{\mathbb{C}}\{h, g, v\}$ is stable under S and T, and therefore under the full modular group. Linear independence of these three functions is not assumed or needed.

The formal matrices for (3), with functions as a column, are

$$A_\varepsilon = \begin{pmatrix} \varepsilon & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad B_\varepsilon = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & -\varepsilon \end{pmatrix}. \tag{4}$$

They satisfy $A_\varepsilon^2 = (A_\varepsilon B_\varepsilon)^3 = B_\varepsilon^6 = I$.

If $\varepsilon = 1$, the table gives $T^3 = -\mathrm{Id}$ on all of V (equivalently, $B_1^3 = -I$). Lemma 3 immediately implies $h|\Gamma(6) = h$. This part has used only hypotheses (1) and (2).

Suppose now that $\varepsilon = -1$. Put

$$u = g + v. \tag{5}$$

The table gives

$$u|S = u, \qquad u|T = u. \tag{6}$$

Thus u is invariant under the full modular group. We have not yet proved $u = 0$, and its invariance alone would not suffice. The next section uses the Euler condition at 5 to prove precisely this.

4 The $m = 5$ relation kills the defect

We record the needed prime relation explicitly. Hypothesis (3) is equivalent to

$$f|\mathcal{T}_5 = 5^{1-k/2}a_5 f, \qquad \mathcal{T}_5 = \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix} + \sum_{r=0}^{4} \begin{pmatrix} 1 & r \\ 0 & 5 \end{pmatrix}.$$

Conjugate this equation by H_{36}, using (1), and subtract the two equations. The two diagonal terms cancel. Right-multiplication by $\mathrm{diag}(5,1)$ then gives

$$f\Big|\sum_{r=1}^{4}\beta(r/5) = f\Big|\sum_{c=1}^{4}\begin{pmatrix}25 & 0\\ -180c & 1\end{pmatrix}. \tag{7}$$

We also know $f|W_{36} = f$ for $W_{36} = \begin{pmatrix}1 & 0\\ 36 & 1\end{pmatrix}$, by periodicity and (1). Changing c by a multiple of 5 on the right-hand side of (7) is therefore permitted by left multiplication with a power of W_{36}. Reindex c by $36bc \equiv -1 \pmod 5$, and set

$$\gamma_b = \begin{pmatrix}5 & -b\\ -36c & (1+36bc)/5\end{pmatrix}.$$

Since $\gamma_b\beta(b/5) = \begin{pmatrix}5 & 0\\ -36c & 1/5\end{pmatrix}$, we obtain the Conrey–Farmer prime relation

$$f\Big|\sum_{b\in\{1,-1,2,-2\}}(1-\gamma_b)\beta(b/5) = 0. \tag{8}$$

This is the $m = 5$ instance of [1, Corollary 2]; the derivation above also checks the normalizations and the permitted left replacements.

Choose explicitly

$$\gamma_1 = \begin{pmatrix}5 & -1\\ 36 & -7\end{pmatrix},\quad \gamma_{-1} = \begin{pmatrix}5 & 1\\ -36 & -7\end{pmatrix},\quad \gamma_2 = \begin{pmatrix}5 & -2\\ -72 & 29\end{pmatrix},\quad \gamma_{-2} = \begin{pmatrix}5 & 2\\ 72 & 29\end{pmatrix}.$$

For $D = \mathrm{diag}(1,6)$ put $\delta_b = D^{-1}\gamma_b D$. Conjugating (8) gives

$$h\Big|\sum_b(1-\delta_b)\beta(6b/5) = 0. \tag{9}$$

The four integral matrices are

$$\delta_1 = \begin{pmatrix}5 & -6\\ 6 & -7\end{pmatrix},\quad \delta_{-1} = \begin{pmatrix}5 & 6\\ -6 & -7\end{pmatrix},\quad \delta_2 = \begin{pmatrix}5 & -12\\ -12 & 29\end{pmatrix},\quad \delta_{-2} = \begin{pmatrix}5 & 12\\ 12 & 29\end{pmatrix}.$$

The following are exact matrix identities:

$$\delta_{\pm1} = T^{\pm1}ST^{\pm6}ST^{\mp1}, \tag{10}$$

$$\delta_{\pm2} = (ST^{\pm2}ST^{\mp2})^2. \tag{11}$$

Continue to assume $\varepsilon = -1$. Since $T^6 = 1$ on V, identity (10) gives $h|\delta_{\pm1} = h$. For the other two, (3)–(6) give

$$h|ST^2ST^{-2} = h - u, \qquad h|ST^{-2}ST^2 = h + u.$$

The function u is invariant under both words. Squaring them proves

$$h|\delta_2 = h - 2u, \qquad h|\delta_{-2} = h + 2u. \tag{12}$$

Thus (9) reduces to

$$2u|\beta(12/5) - 2u|\beta(-12/5) = 0.$$

Consequently $u|\beta(24/5) = u$, and $u|T = u$ then implies $u|\beta(1/5) = u$. Together with $u|S = u$, this gives

$$u|E = u, \qquad E = S\beta(1/5) = \begin{pmatrix} 0 & -1 \\ 1 & 1/5 \end{pmatrix}.$$

Here $\det E = 1$ and $\operatorname{tr} E = 1/5$. The matrix is elliptic and has infinite order: a finite-order matrix has algebraic-integer trace, whereas the rational number $1/5$ is not an algebraic integer. Lemma 2 proves $u = 0$.

It follows that $v = -g$, so $V = \operatorname{span}\{h, g\}$, with

$$h|S = -h, \quad g|S = -g, \quad h|T = g, \quad g|T = g - h.$$

Again $T^3 = -\mathrm{Id}$ on V. Lemma 3 proves $h|\Gamma(6) = h$ in this sign as well.

5 Cuspidality and descent to level 36

We have established, for either sign, that h is $\Gamma(6)$-invariant, that $V = \operatorname{span}\{h|\sigma : \sigma \in \mathrm{SL}_2(\mathbb{Z})\}$ is finite dimensional, and that $T^3 = -\mathrm{Id}$ on V. For every $\sigma \in \mathrm{SL}_2(\mathbb{Z})$, this gives

$$(h|\sigma)(z + 3) = -(h|\sigma)(z). \tag{13}$$

We justify cusp holomorphy without adding a Ramanujan-type estimate. From $a_n = O(n^C)$, uniformly in x,

$$|f(x + iy)| \ll y^{-C-1} \quad (0 < y \le 1), \qquad |f(x + iy)| \ll e^{-\pi y} \quad (y \ge 1).$$

The first bound follows by comparison of $\sum n^C e^{-2\pi ny}$ with its defining integral. If the lower-left entry of σ is nonzero, then on a bounded horizontal strip as $y \to \infty$, the imaginary part of $\sigma(x + iy)/6$ is comparable to y^{-1} and the slash factor is $O(y^{-k})$. Thus $h|\sigma$ grows at most polynomially at infinity. If that entry is zero, it decays exponentially.

Each $h|\sigma$ is holomorphic and 6-periodic, so it is a holomorphic function of $q_6 = e^{2\pi iz/6}$ on a punctured disk. Polynomial growth in y rules out all negative Laurent coefficients: on $|q_6| = r$ the bound is $O((1 + |\log r|)^A)$ for some A, and Cauchy's formula bounds the coefficient of q_6^{-j} by $O(r^j(1+|\log r|)^A) \to 0$ for $j > 0$. The constant coefficient vanishes by (13). Therefore $h \in S_k(\Gamma(6))$.

Finally, for $\gamma = \begin{pmatrix} a & b \\ 36c & d \end{pmatrix} \in \Gamma_0(36)$,

$$D^{-1}\gamma D = \begin{pmatrix} a & 6b \\ 6c & d \end{pmatrix}.$$

Because $ad \equiv 1 \pmod 6$ and $(\mathbb{Z}/6\mathbb{Z})^\times = \{1, -1\}$, this belongs to $\Gamma(6)$ or $-\Gamma(6)$. Conversely every matrix in $\Gamma(6) \cup -\Gamma(6)$ is obtained in this way. Thus

$$D^{-1}\Gamma_0(36)D = \Gamma(6) \cup -\Gamma(6).$$

Since k is even and $h = 6^{k/2} f|D$, transformation laws and cusp vanishing transfer under this conjugation. We obtain $f \in S_k(\Gamma_0(36))$, completing the proof of Theorem 1.

6 Verification

The accompanying file `verify_N36.py` uses only exact integer and rational arithmetic from the Python standard library. It verifies the four matrices, their word factorizations, the representation relations, the two defect formulas, and the conjugation of the prime relation. Independently of Lemma 3, it verifies both final representations on all 72 elements of $\mathrm{SL}_2(\mathbb{Z}/6\mathbb{Z})/\{\pm I\}$, checking both generator edges at every element. These computations check the algebra; the analytic vanishing and cusp arguments are proved above.

References

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[3] B. Bedert, G. Cooper, T. Oliver, and P. Zhang, *Twisting moduli for* GL(2), Journal of Number Theory 217 (2020), 142–162. arXiv:2003.02557.